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MHHS2402 CARDIOVASCULAR AND RESPIRATORY SYSTEM.

| Block 1: Respiratory System

Week 4: Respiratory regulation inputs

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Macquarie University. On the land of the Wallumattagal clan of the Dharug Nation.

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The unit staff are here to get you through this unit (not to fail you).



LEARNING OBJECTIVE

Ventilation-perfusion control systems

Recognise the role of control systems to maintain appropriate ventilation and perfusion for homeostasis.



| Homeostasis

Homeostasis is a **dynamic** (always changing) process where a biological system maintains certain variables (e.g. temperature, pH, blood flow to an organ) at an optimal level.

It is about **balance**, not about being “at rest”. For example, dynamic changes are needed to maintain:

- blood pH homeostasis during exercise;
- blood glucose homeostasis after eating;
-  body temperature homeostasis on a very hot day.

| Homeostasis

FOR A HOMEOSTATIC REGULATION TO OCCUR...

1. **Change:** Something in the body or surrounding environment must **change**.
2. **Detect:** Something in the body must be able to **detect** that change.
3. **Process:** Something in the body must **process** the signal received from the detector.
4. **Action:** The decision from the processing is sent to the respiratory muscles to change minute ventilation.



Respiratory rate and tidal volume do not change because we "think" they should.

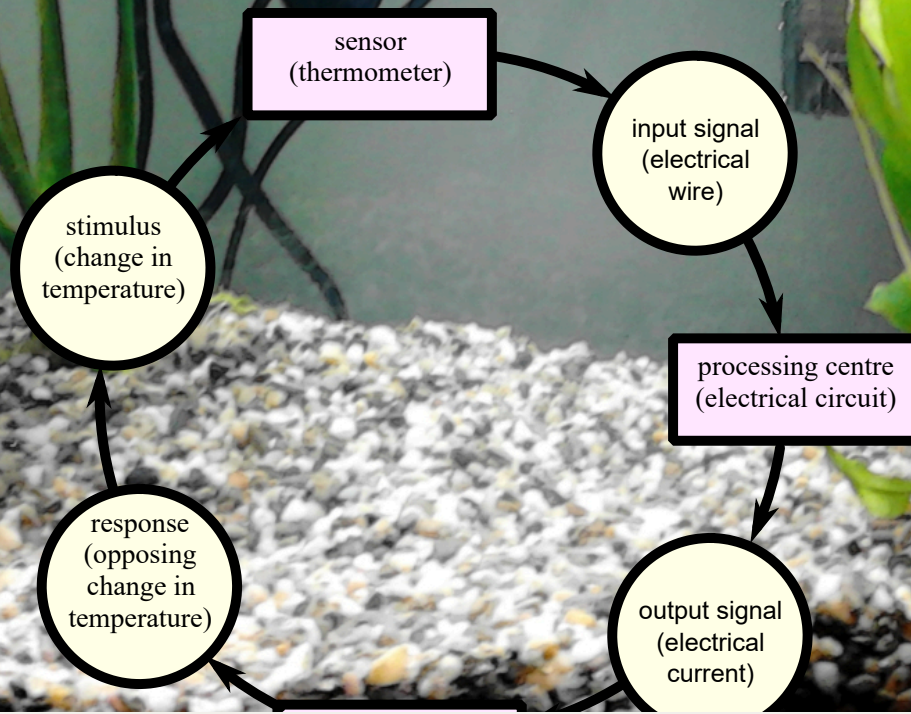
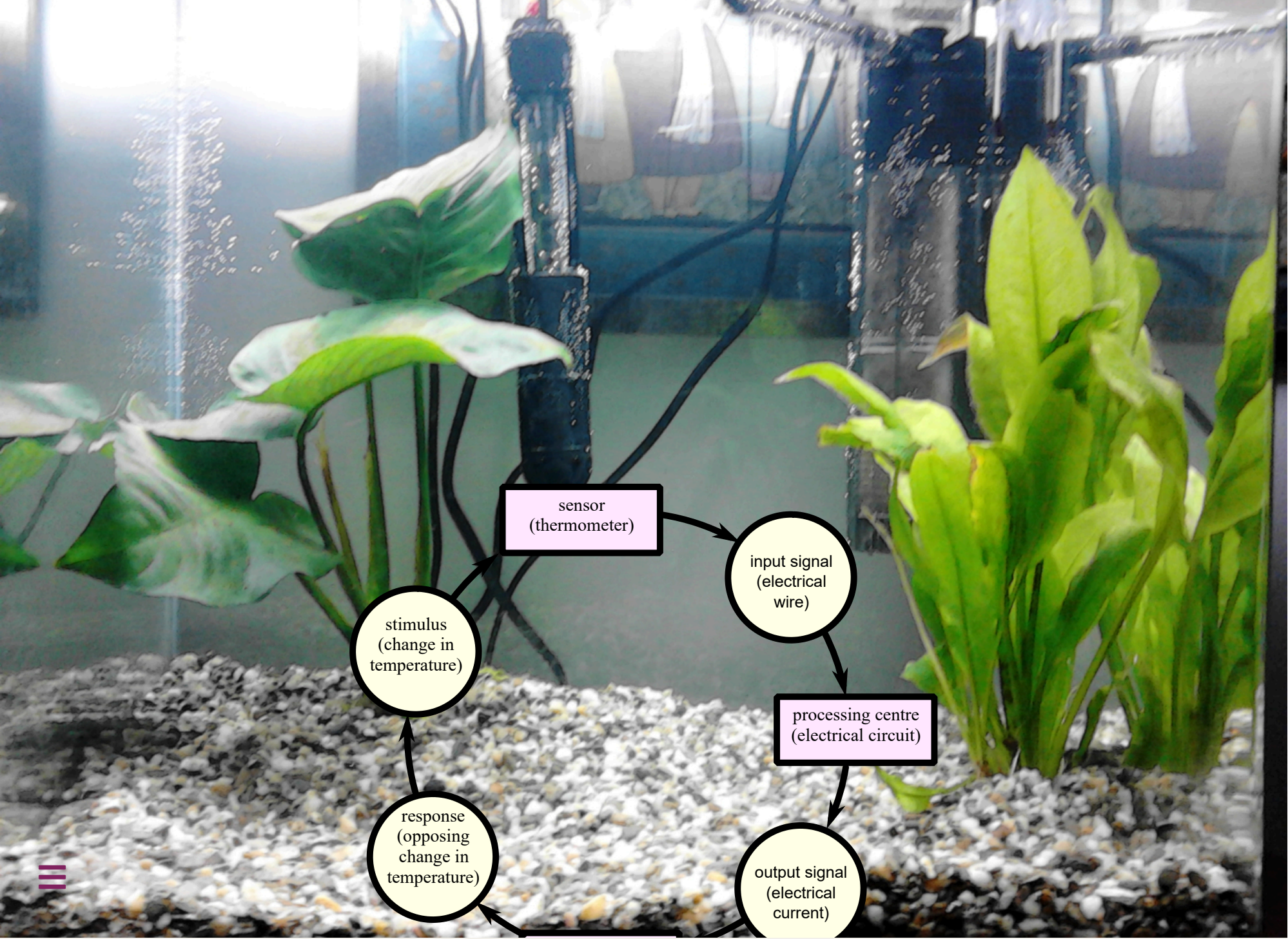


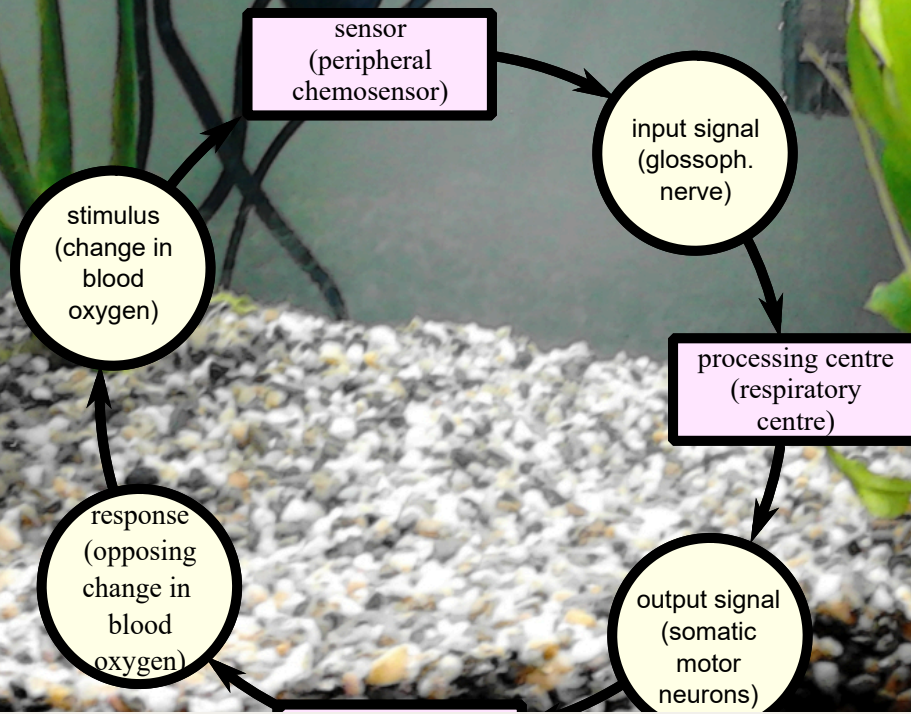
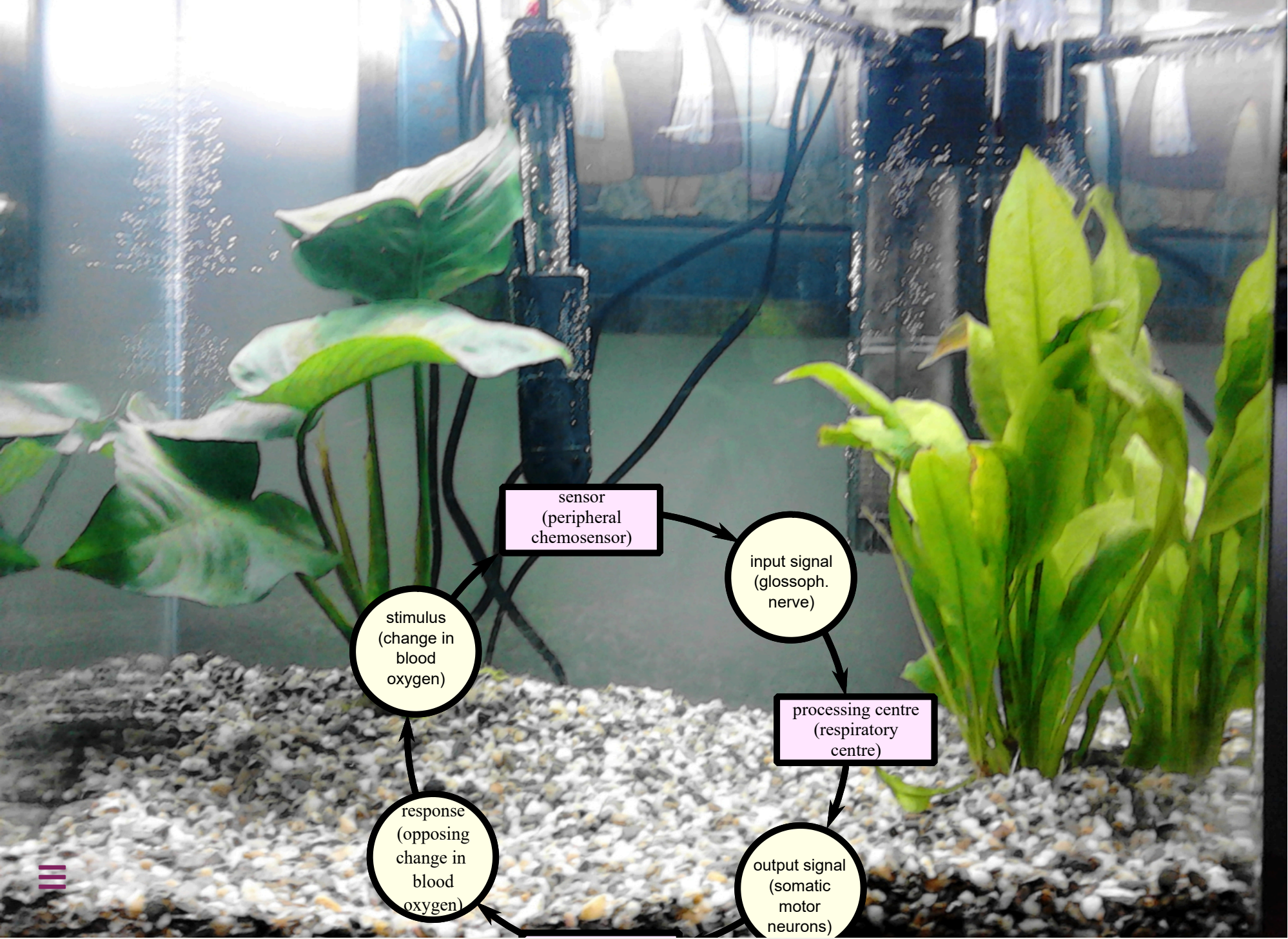
If you think tidal volume or respiratory rate changed, first answer:

What changed in the environmental or in the body first

→ How was this detected?

⇒ Where did that signal get sent and how was it processed?





| Homeostasis

NEGATIVE FEEDBACK

The level of something in the body **increases**.

→ This is detected, processed, and a signal fed back that...

→ **decreases** that initial thing.

or



| Homeostasis

NEGATIVE FEEDBACK

Minute ventilation **increases**.

→ This is detected, processed, and a signal fed back that...

→ **decreases** minute ventilation.

or





| Homeostasis
NEGATIVE FEEDBACK. EXAMPLE.

| Homeostasis

POSITIVE FEEDBACK

The level of something in the body **increases**.

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or



| Homeostasis

POSITIVE FEEDBACK

Minute ventilation **increases**.

→ This is detected, processed, and a signal fed back that...

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Homeostasis 19
POSITIVE FEEDBACK. EXAMPLE.

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| Reflection quiz

VENTILATION-PERFUSION CONTROL SYSTEMS

RECOGNISE THE ROLE OF CONTROL SYSTEMS TO MAINTAIN APPROPRIATE VENTILATION AND PERFUSION FOR HOMEOSTASIS.

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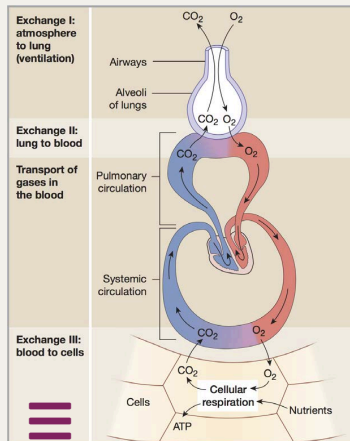
LEARNING OBJECTIVE

Arterial Blood Parameters

Know the normal levels of the three key arterial blood parameters used to regulate ventilation and perfusion: oxygen, carbon dioxide, and pH.

Arterial blood gases and pH

WHY GASES IN BLOOD – THIS IS A RESPIRATORY LECTURE?



Gas exchange occurs from an area of higher gas partial to pressure to an area of lower gas partial pressure.

| Arterial blood gases and pH

WHY GASES IN BLOOD – THIS IS A RESPIRATORY LECTURE?

Partial pressures of oxygen and carbon dioxide in the body.

	alveoli		venous blood	arterial blood		tissue cells	venous blood
P_{O_2} (%)	100	→	40	85 - 100	→	10 - 30	40
P_{CO_2} (%)	40	←	46	35 - 45		50	→ 46
			↓	↓		↓	↓
pH			7.37	7.38 - 7.42		7.34	7.37



| Arterial blood gases and pH

CARBON DIOXIDE IN BLOOD

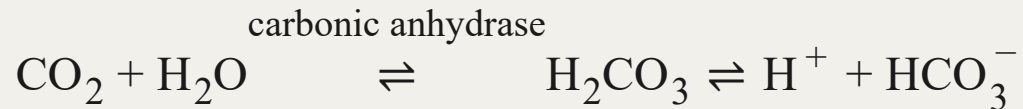
Dissolving of carbon dioxide in blood (water):



| Arterial blood gases and pH

CARBON DIOXIDE IN BLOOD

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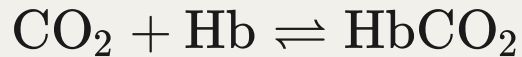
carbon dioxide + water $\rightleftharpoons$ carbonic acid $\rightleftharpoons$ hydrogen ions + bicarbonate ions



| Arterial blood gases and pH

CARBON DIOXIDE IN BLOOD

Carriage of carbon dioxide by haemoglobin:



carbon dioxide + haemoglobin $\rightleftharpoons$ carbaminohaemoglobin



| Arterial blood gases and pH

OXYGEN IN BLOOD

Oxygen is carried in two ways in the blood:

1. bound to haemoglobin
2. dissolved in plasma



| Arterial blood gases and pH

We have established that:

- oxygen is important for cellular respiration
- carbon dioxide is a by-product of cellular respiration and needs to be removed
- carbon dioxide decreases pH¹

Effective functioning of body processes require rather narrow ranges of these parameters, *especially pH.*



Arterial blood gases and pH

NORMAL RANGES OF ARTERIAL BLOOD GASES, PH AND BICARBONATE

arterial blood	low	normal	high
paO ₂	< 80 mmHg	80 -- 100 mmHg	> 100 mmHg
O ₂ saturation	< 95 %	95 - 99 %	N/A
paCO ₂	< 35 mmHg	35 - 45 mmHg	> 45 mmHg
pH	< 7.35 (acidosis)	7.35 - 7.45	> 7.45 (alkalosis)
HCO ₃ ⁻	< 22 mEq/L	22 - 26 mEq/L	> 26 mEq/L



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Reflection quiz

ARTERIAL BLOOD PARAMETERS

KNOW THE NORMAL LEVELS OF THE THREE KEY ARTERIAL BLOOD PARAMETERS USED TO REGULATE VENTILATION AND PERFUSION: OXYGEN, CARBON DIOXIDE, AND PH.

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LEARNING OBJECTIVE

Chemoreceptor function

Identify how central and peripheral chemoreceptors monitor carbon dioxide and oxygen levels to regulate breathing.

| Chemoreceptor input to the respiratory centre

Note: In the context of this learning outcome we are discussing chemoreceptor feedback to breathing regulation.

Chemoreceptors also feedback to the cardiovascular centres of the brain. More on this in Week 6.



| Chemoreceptor input to the respiratory centre

PERIPHERAL CHEMORECEPTORS



| Chemoreceptor input to the respiratory centre

PERIPHERAL CHEMORECEPTORS



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| Chemoreceptor input to the respiratory centre

CENTRAL CHEMORECEPTORS

Centres in the brain that regulate breathing, in addition to receiving signals from the peripheral chemoreceptors, are *directly* chemosensitive.



Then how do the central chemoreceptors modify ventilation in response to partial pressures of blood gases?



| Chemoreceptor input to the respiratory centre

CENTRAL CHEMORECEPTORS



carbon dioxide + water $\rightleftharpoons$ carbonic acid $\rightleftharpoons$ hydrogen ions + bicarbonate ions



Central chemoreceptors are, therefore, highly sensitive to blood CO_2 , but not blood H^+ .





| Chemoreceptor input to the respiratory centre

A PATHWAY OF HOMEOSTASIS



| Chemoreceptor input to the respiratory centre
A PATHWAY OF HOMEOSTASIS



| Chemoreceptor input to the respiratory centre

A PATHWAY OF HOMEOSTASIS

| Reflection quiz

CHEMORECEPTOR FUNCTION

IDENTIFY HOW CENTRAL AND PERIPHERAL CHEMORECEPTORS MONITOR CARBON DIOXIDE AND OXYGEN LEVELS TO REGULATE BREATHING.

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LEARNING OBJECTIVE

Respiratory reflexes and receptors

Understand the various receptors and reflexes affecting respiratory centers, including stretch (Hering-Breuer),



Reflex inputs to the respiratory centre

reflex

The sum total of any particular involuntary activity.

Dorland's Illustrated Medical Dictionary, 29th Ed.



What are some reflexes that would influence breathing / ventilation?



| Reflex inputs to the respiratory centre

HERING-BREUER REFLEX

- **Stretch receptors** are located in the bronchi and bronchioles.
- At high levels of stretch, they communicate to the central respiratory centre via the vagi to **cease inspiration**.
- **Only activated during hyperinflation** (tidal volume $> 3 \times$ normal tidal volume).



| Reflex inputs to the respiratory centre

APNOEA

The cessation of breathing for a period of time.

Apnoea can be:



| Reflex inputs to the respiratory centre

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| Reflex inputs to the respiratory centre

INCREASED VENTILATION

Occurs under a number of scenarios. There are at least four different receptors that can increase minute ventilation, and may lead to hyperventilation:



| Reflex inputs to the respiratory centre

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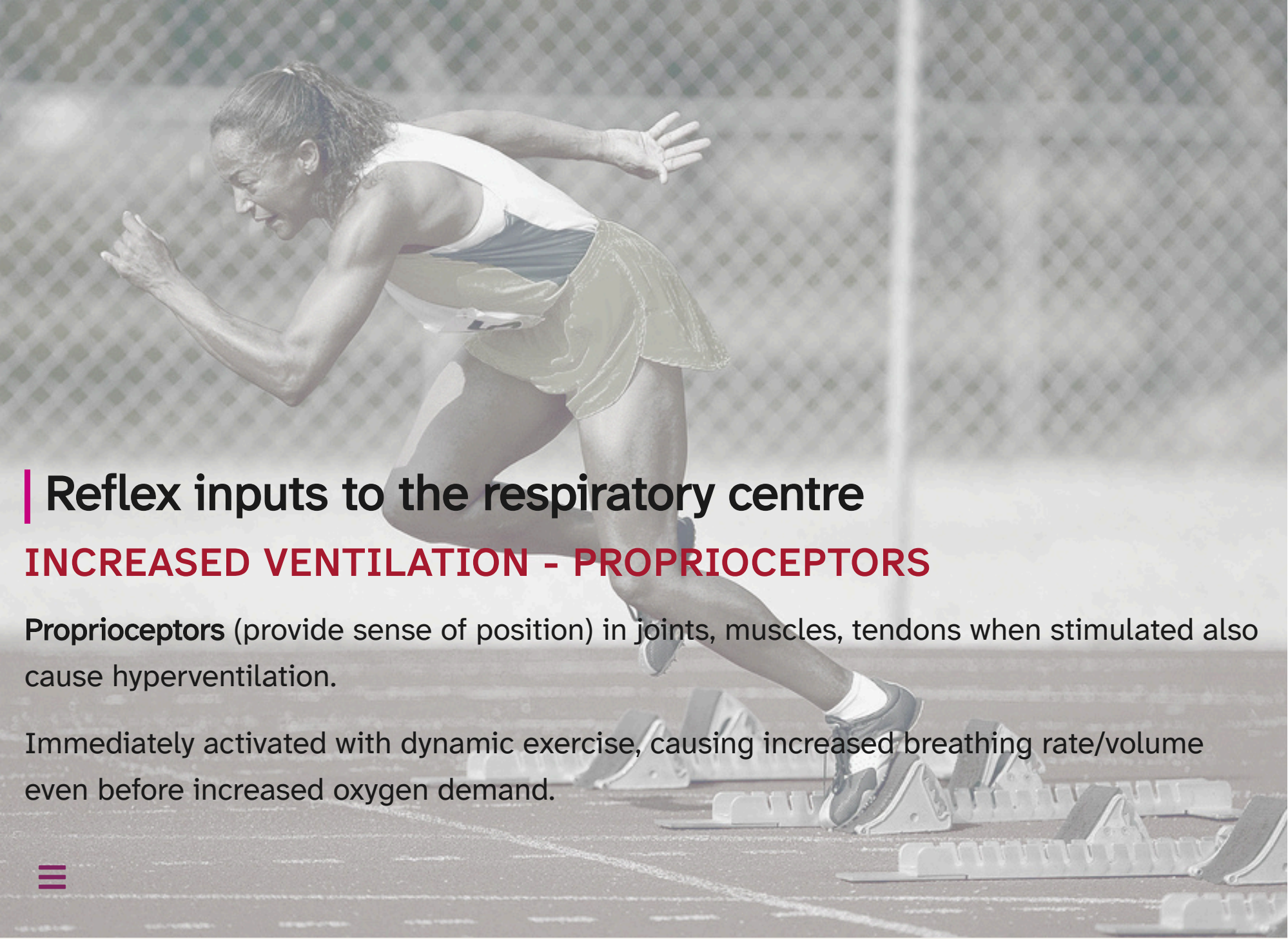
| Reflex inputs to the respiratory centre

INCREASED VENTILATION - IRRITANT RECEPTORS

Irritant receptors in the lungs stimulated by harmful chemicals (e.g. ammonia, sulphur dioxide).

Irritant receptor stimulation results in reflex hyperventilation and bronchospasm, reducing likelihood of harmful chemicals entering the alveoli.





| Reflex inputs to the respiratory centre

INCREASED VENTILATION - PROPRIOCEPTORS

Proprioceptors (provide sense of position) in joints, muscles, tendons when stimulated also cause hyperventilation.

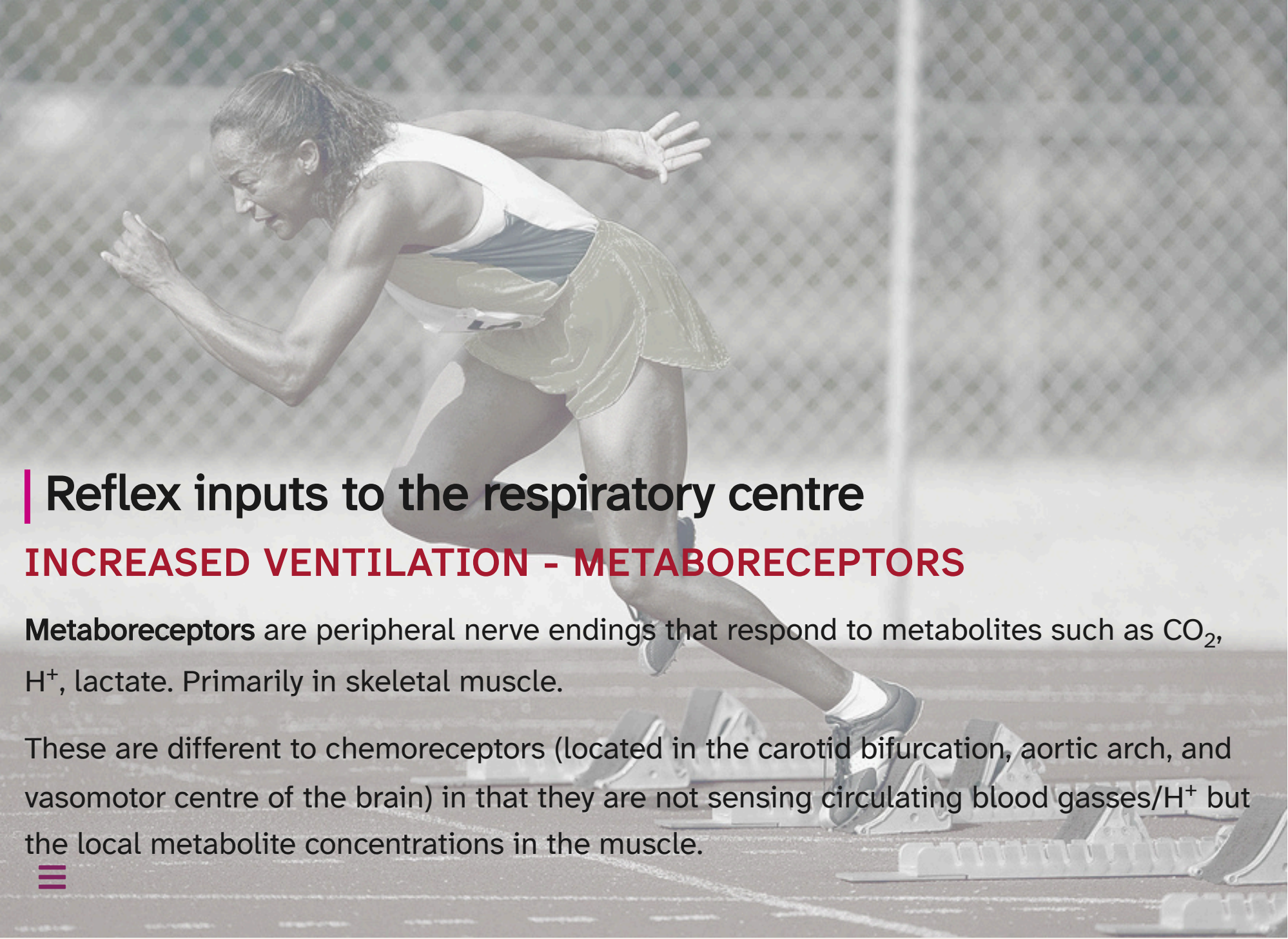
Immediately activated with dynamic exercise, causing increased breathing rate/volume even before increased oxygen demand.



| Reflex inputs to the respiratory centre

INCREASED VENTILATION - METABORECEPTORS

Metaboreceptors are peripheral nerve endings that respond to metabolites such as CO_2 , H^+ , lactate. Primarily in skeletal muscle.



Reflex inputs to the respiratory centre

INCREASED VENTILATION - METABORECEPTORS

Metaboreceptors are peripheral nerve endings that respond to metabolites such as CO_2 , H^+ , lactate. Primarily in skeletal muscle.

These are different to chemoreceptors (located in the carotid bifurcation, aortic arch, and vasomotor centre of the brain) in that they are not sensing circulating blood gasses/ H^+ but the local metabolite concentrations in the muscle.





| Reflex inputs to the respiratory centre

INCREASED VENTILATION - THERMORECEPTORS

Thermoreceptors, mostly cutaneous, responding to body temperature outside of the normal range. Both increased *and* decreased body temperature can cause increased ventilation.



Reflex inputs to the respiratory centre

INCREASED VENTILATION - PAIN RECEPTORS

Pain receptors also signal to the respiratory centre.

Also can cause a hyperventillatory response.



Reflex inputs to the respiratory centre

COUGH REFLEX





| Reflex inputs to the respiratory centre

SNEEZE REFLEX

Similar reflex to a cough, but initiated by irritation of the nasal mucus membrane.

| Reflex inputs to the respiratory centre

DEGLUTATION REFLEX

- During swallowing of the food, respiration is arrested.
- This is a type of apnoea (aka swallowing apnoea or deglutation apnoea).
- Prevents entry of the food particles into the respiratory tract.
- During the swallow, the hyoid bone can be seen elevating, and the epiglottis closing as a result.



Reflex inputs to the respiratory centre

RESPIRATORY RECEPTORS AND REFLEXES: SUMMARY

receptor	located	senses	reflex
central chemoreceptors	central (brain)	blood O ₂ , CO ₂	alters minute ventilation
peripheral chemoreceptors	peripheral (carotid/aorta)	blood O ₂ , CO ₂ , H ⁺	alters minute ventilation
juxta-capillary receptors	alveoli	congestion	apnoea
lung irritant receptors	respiratory epithelium	airborne irritants	cough, sneeze
pain receptors	skin, some internal tissue	pain	increased minute ventilation
proprioceptors	throughout body	movement	increased minute ventilation
metaboreceptors	primarily skeletal muscle	metabolite concentration	increased minute ventilation

Reflex inputs to the respiratory centre

RESPIRATORY RECEPTORS AND REFLEXES: SUMMARY

reflex	receptor	action
cough	respiratory irritant receptors	increased intrathoracic pressure and release
sneeze	respiratory irritant receptors	increased intrathoracic pressure and release
deglutition	pharynx touch receptors	apnoea
Hering-Breuer	bronchi/bronchiole stretch receptors	cessation of inspiration
hyperventilation*	lung irritant; proprioceptors; thermoreceptors; pain receptors	increased minute ventilation

| Reflex inputs to the respiratory centre

RESPIRATORY RECEPTORS AND REFLEXES: SUMMARY



| Reflection quiz

RESPIRATORY REFLEXES AND RECEPTORS

UNDERSTAND THE VARIOUS RECEPTORS AND REFLEXES AFFECTING RESPIRATORY CENTERS, INCLUDING STRETCH (HERING-BREUER), JUXTACAPILLARY, IRRITANT, PROPRIOCEPTORS, THERMORECEPTORS, PAIN, COUGH, SNEEZING, AND DEGLUTITION.

LEARNING OBJECTIVE

Respiratory centre regulation

Recognise the role of the respiratory center in regulating respiration.



| The respiratory centre

To have an **effect**, the outcome of the respiratory input has to alter minute ventilation.

This means all respiratory regulatory responses have to illicit a change in either or both:

- breathing rate
- tidal volume

$$\text{minute ventilation} = \text{ventilation rate} \times \text{tidal volume}$$



The respiratory centre



The respiratory centre



The respiratory centre



The respiratory centre



The respiratory centre



The respiratory centre



| The respiratory centre

INPUTS TO THE RESPIRATORY CENTRE

The respiratory centre of the brain sets the respiratory rate and respiratory depth (i.e. minute ventilation) in response to the inputs it receives. These inputs include:

- **higher order brain centres** responding to environmental stressors.
- **peripheral chemoreceptors** signal received by the glossopharyngeal nerve.
- **central chemoreceptors** located in the intracranial space
-  **other sensory nerves** (see previous lecture).

| The respiratory centre

OUTPUTS OF THE RESPIRATORY CENTRE



| The respiratory centre

OUTPUTS OF THE RESPIRATORY CENTRE

Respiratory regulatory responses have to illicit a change in either or both:

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Respiratory regulation: communicating

(as a professional, or... in assessment task 3 or the final exam)

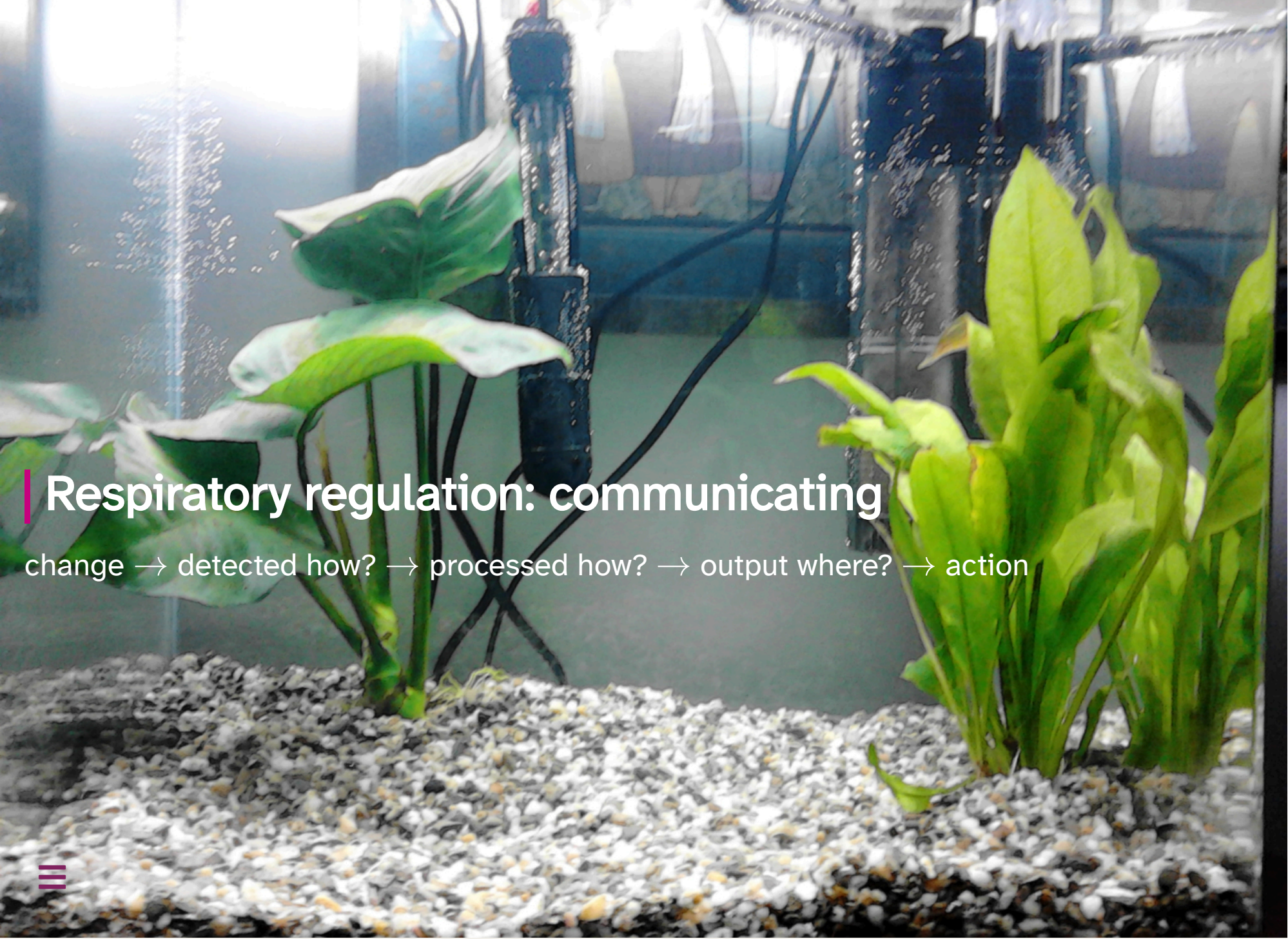


| Respiratory regulation: communicating

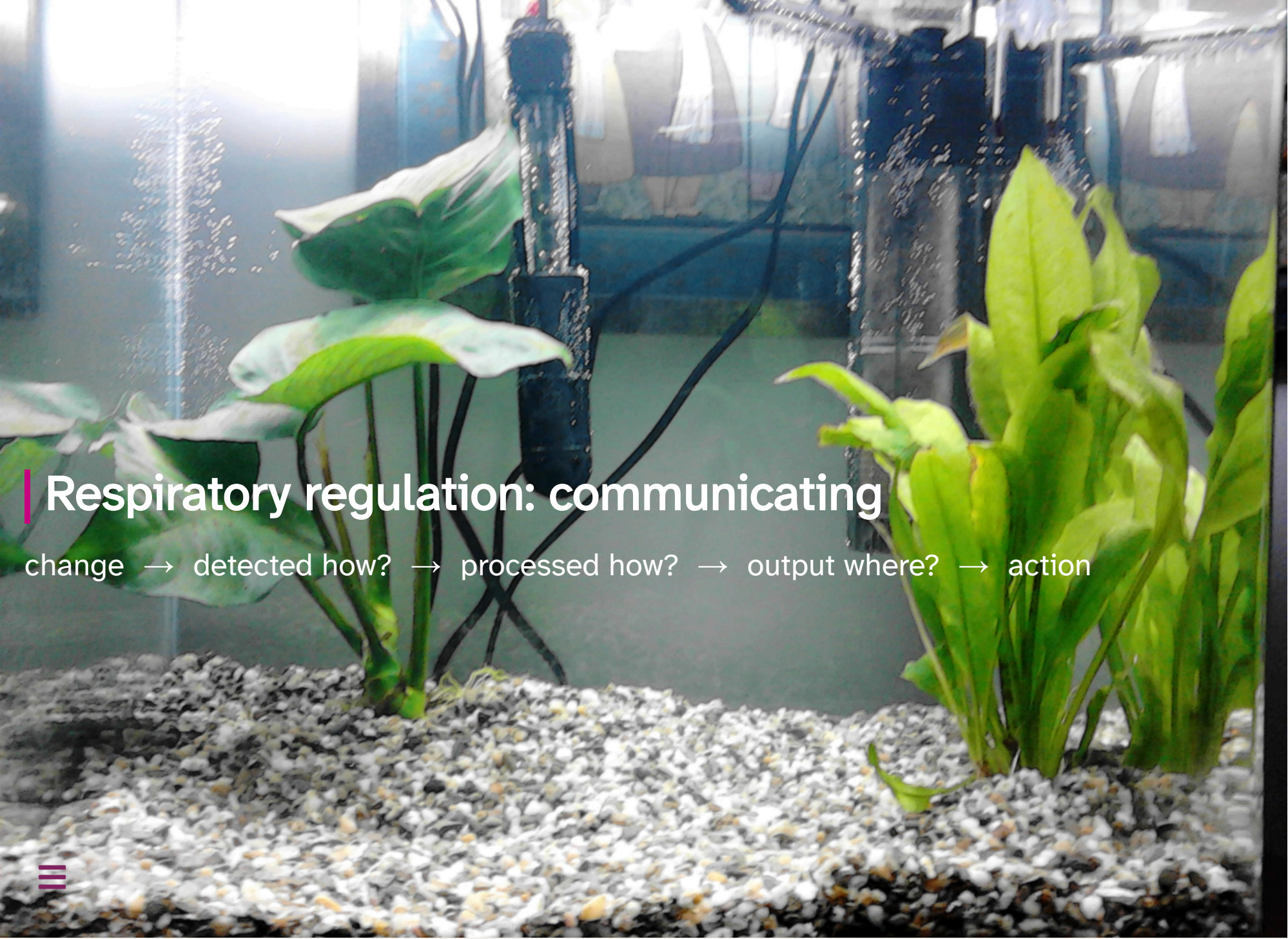
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change → detected how? → processed how? → output where? → action





Respiratory regulation: communicating
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| Respiratory regulation: communicating

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What is one mechanism that increases oxygen intake during walking?

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| Respiratory regulation: communicating

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One of the actions of **walking** is **movement** (change) that **stimulates proprioceptors** (detected). These signal to the **respiratory centre** of the brain (processed), with resulting **increased activity in the phrenic and and intercostal nerves** (output) causing an **increase in breathing rate and tidal volume** (action). This **increases minute ventilation and the partial pressure of oxygen in the alveoli**, which **increases diffusion of oxygen into the blood**.

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07-15 10:11:02!



